Global Olympic Sports Performance Analysis

Project Documentation

Python | pandas | Matplotlib | Seaborn

Abstract

This project analyses Olympic athlete and medal records to understand what drives performance at the Summer and Winter Games. A reproducible Python pipeline cleans the data, answers eight analytical questions and exports charts and summary tables. The version of the dataset bundled with this repository is a simulated sample of 40,147 athlete-event records covering 28 countries, 24 sports and the Games from 1992 to 2020. It follows the schema of the public Kaggle dataset '120 years of Olympic history', so the same code runs unchanged on the real data.

Objectives

1. Rank countries by medals and track the leading nations over time.
2. Measure medal conversion efficiency.
3. Track female participation across Games.
4. Quantify the host-nation effect.
5. Compare age profiles across sports.
6. Show which sports drive each country's medals and the Summer vs Winter split.

Data Note

The bundled dataset is simulated for demonstration and is not official Olympic data. The findings below describe that sample and illustrate the method. To obtain real-world conclusions, place the Kaggle file *athlete_events.csv* in *source code/data/* and re-run *olympic_analysis.py*.

Methodology

Load: the script uses athlete_events.csv when present and falls back to the sample file.
Clean: duplicates are removed, missing medals are labelled 'None', missing ages are filled with the sport median and BMI is derived from height and weight.
Analyse: grouped aggregations in pandas produce medal tables, trends, participation shares, host uplift, age distributions, conversion rates and a country-by-sport matrix.
Visualise: Matplotlib and Seaborn render eight charts at 130 dpi.

1. Medal Table

In the sample, the five leading countries by total medals are United States (568), China (447), Russia (361), Germany (359), Great Britain (286). United States leads with 568 medals.

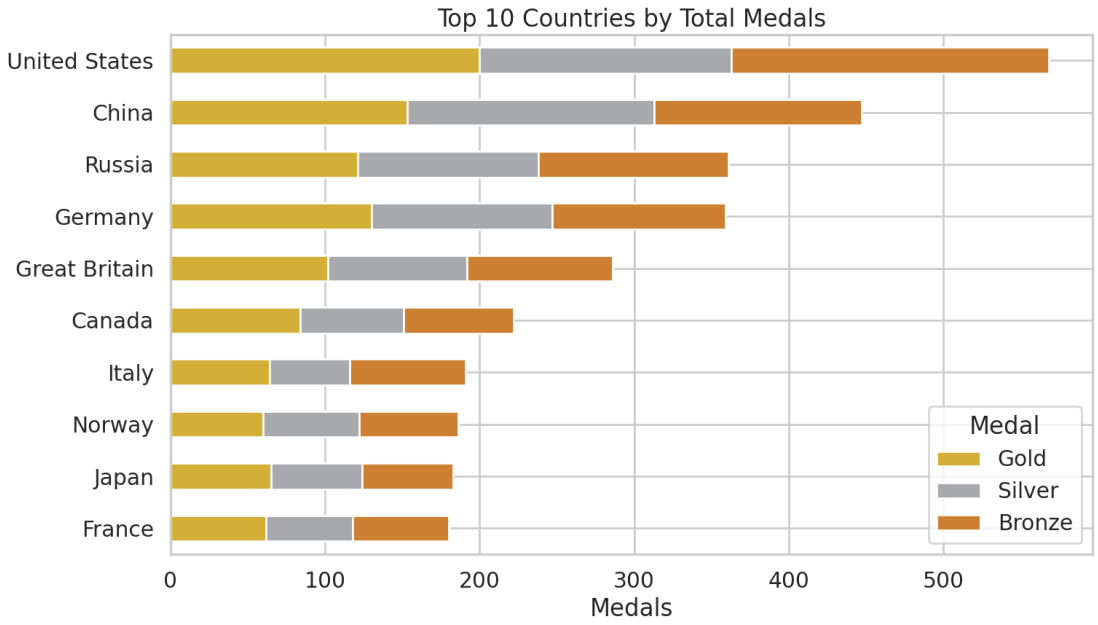


Figure 1: Top 10 countries by gold, silver and bronze medals

2. Medal Trend

The Summer Games trend for the top five countries shows how each nation's medal count moves from one Games to the next. Peaks frequently coincide with a country hosting.

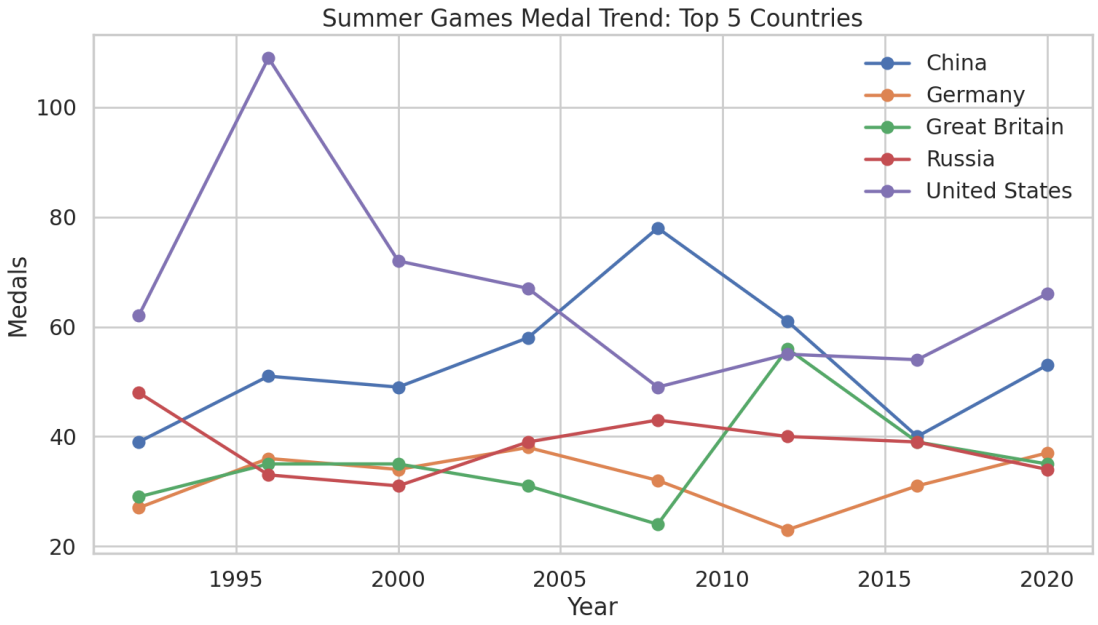


Figure 2: Summer medals per Games, top 5 countries

3. Gender Participation

The female share of athletes moved from 44.6% at the first Games in the sample to 51.7% at the most recent one, reaching parity in the later Games.

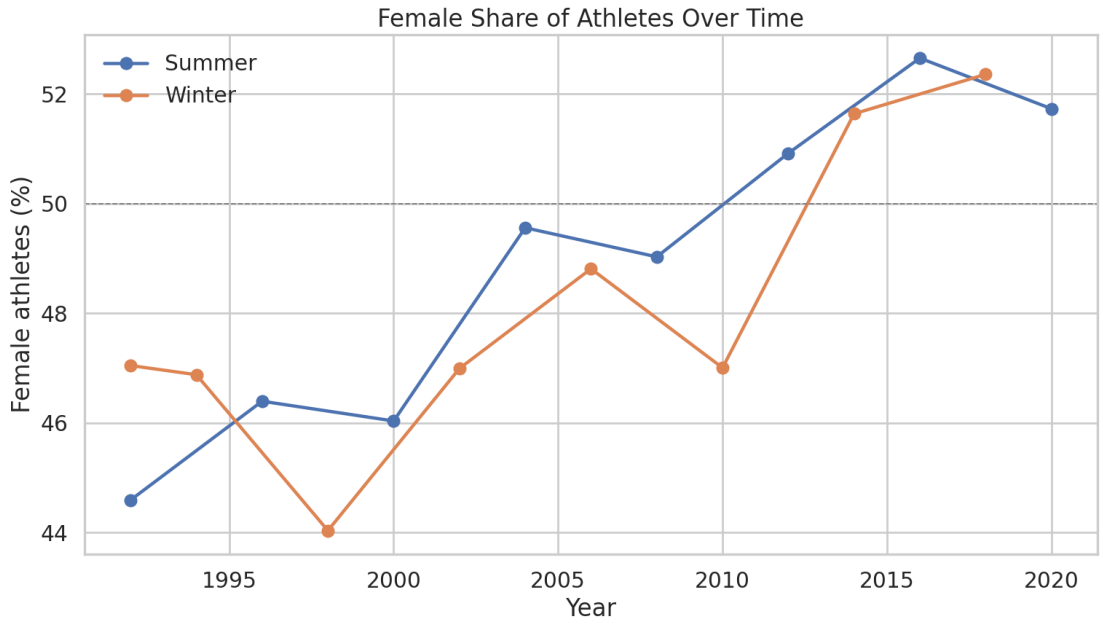


Figure 3: Female share of athletes by Games

4. Host-Nation Effect

Hosts won more medals than their average at other Games in 13 of 15 editions, with a mean uplift of 79.1%. Small-sample countries such as single-appearance winter hosts show the largest swings, so uplift should be read together with absolute medal counts.

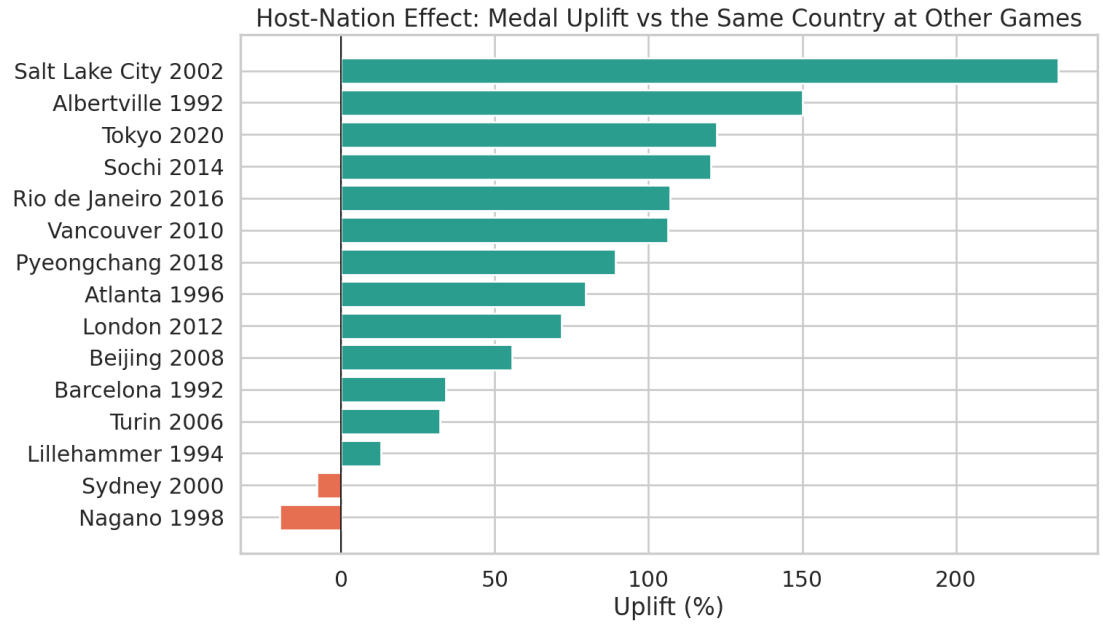


Figure 4: Medal uplift for host nations

5. Age Profile by Sport

Gymnastics has the youngest median age (20) while Shooting has the oldest (33). Technical and precision sports allow longer careers than explosive or flexibility-driven ones.

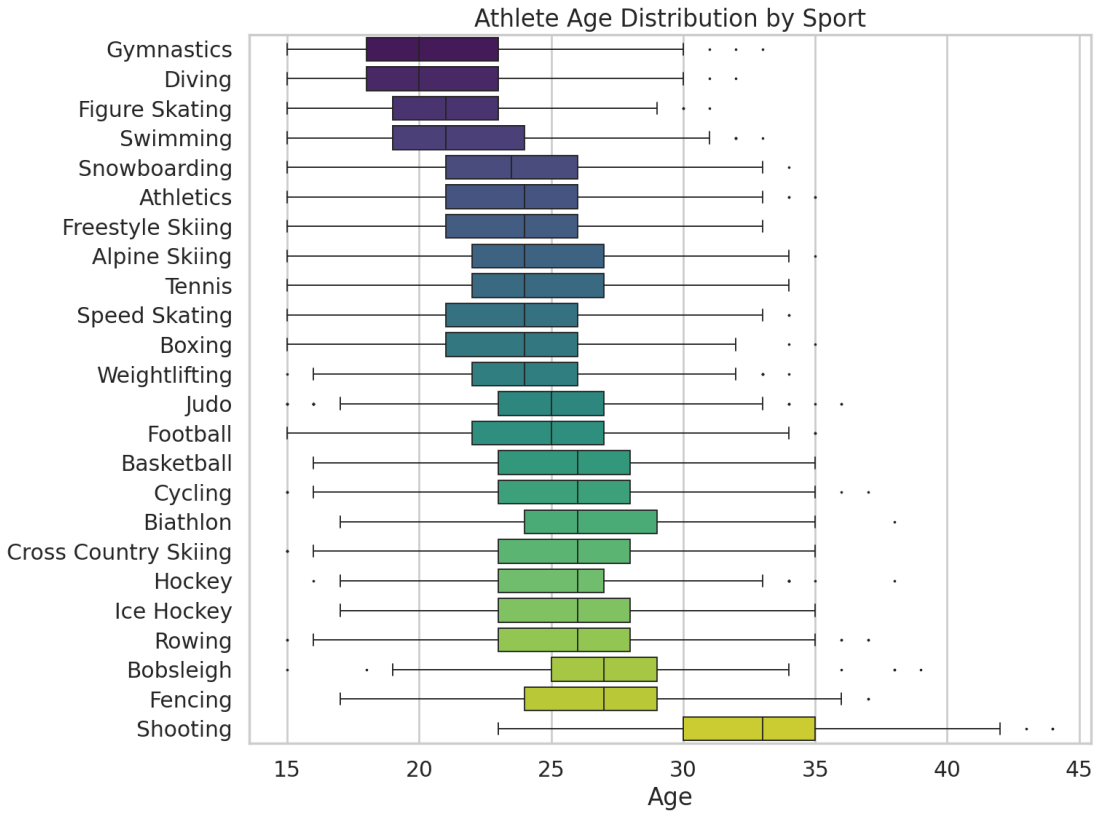


Figure 5: Age distribution by sport

6. Medal Conversion Rate

Conversion rate is medals per 100 athlete entries. United States is highest at 22.86 per 100, showing that depth of talent matters as well as team size.

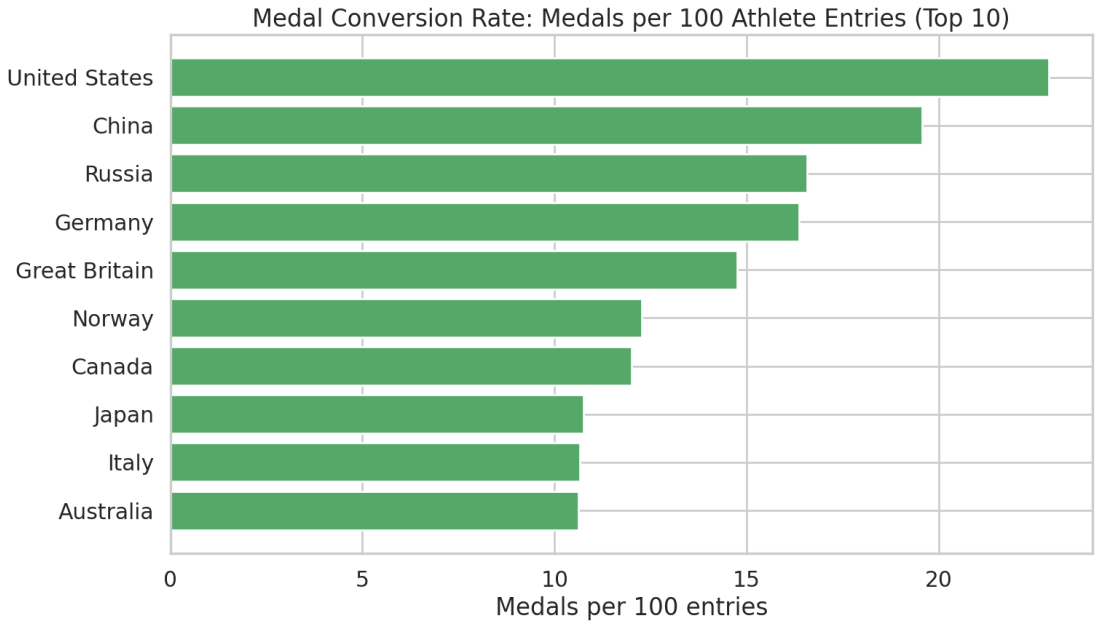


Figure 6: Medals per 100 entries

7. Summer vs Winter

Most leading countries earn the large majority of their medals in the Summer Games, while a few nations (for example Norway and Canada) draw a far bigger share from Winter sports.

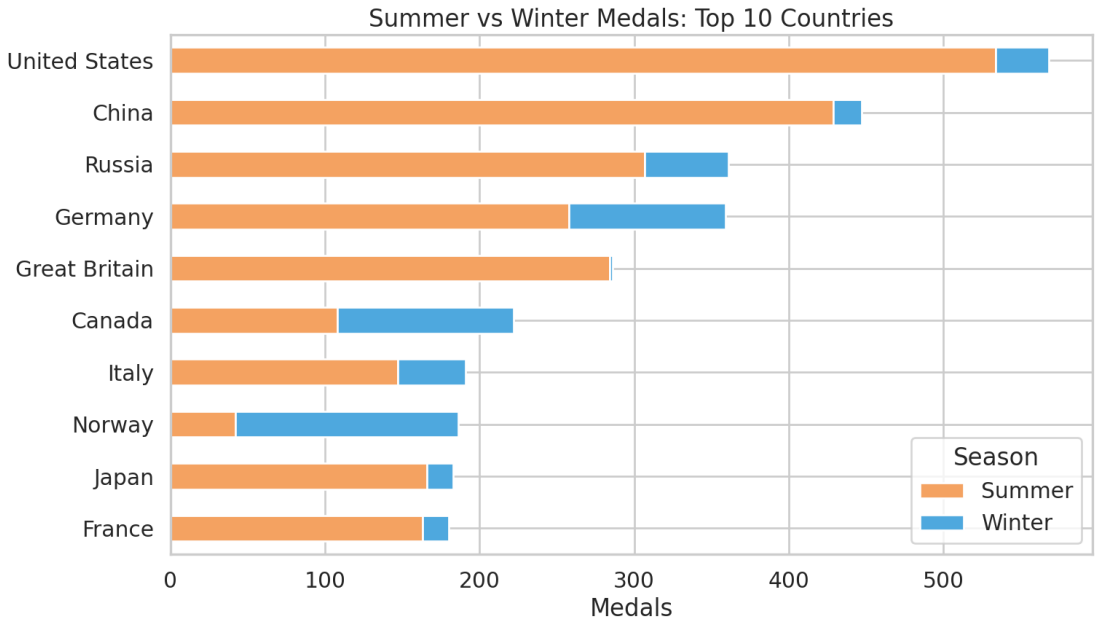


Figure 7: Summer and Winter medals for the top 10

8. Country x Sport Matrix

The heatmap reveals specialisation: Athletics and Swimming dominate the medal counts of the largest teams, while Cross Country Skiing and Alpine Skiing explain the Nordic and Canadian totals.

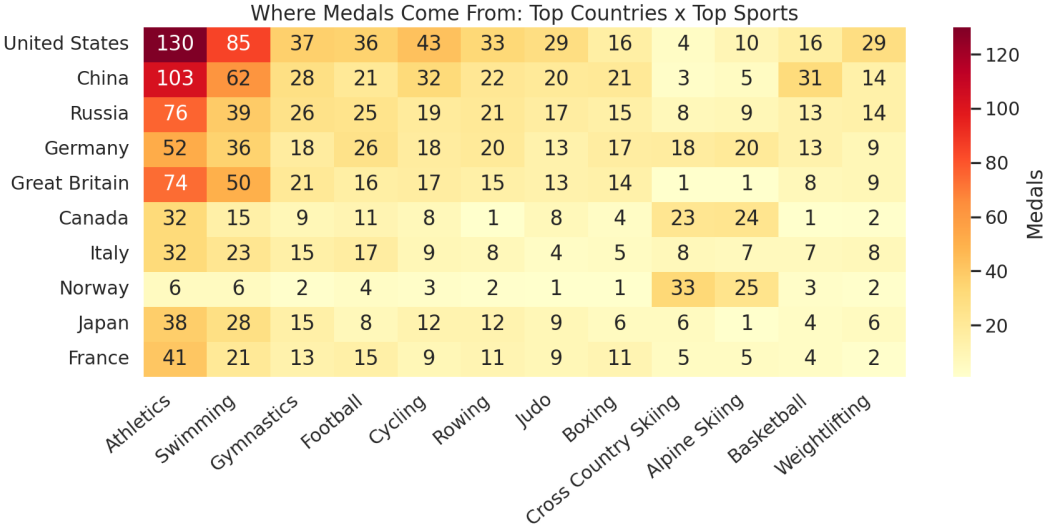


Figure 8: Medals by country and sport

Conclusion

The project demonstrates a complete, reproducible analytics workflow: data loading, cleaning, aggregation, visualisation and reporting. The same pipeline can be pointed at the full 120-year Olympic dataset to produce real findings.

How to Run

cd 'source code'
pip install -r requirements.txt
python olympic_analysis.py

Future Scope

Per-capita and GDP-adjusted comparisons, an interactive Power BI or Tableau dashboard, and a predictive model for medal probability.